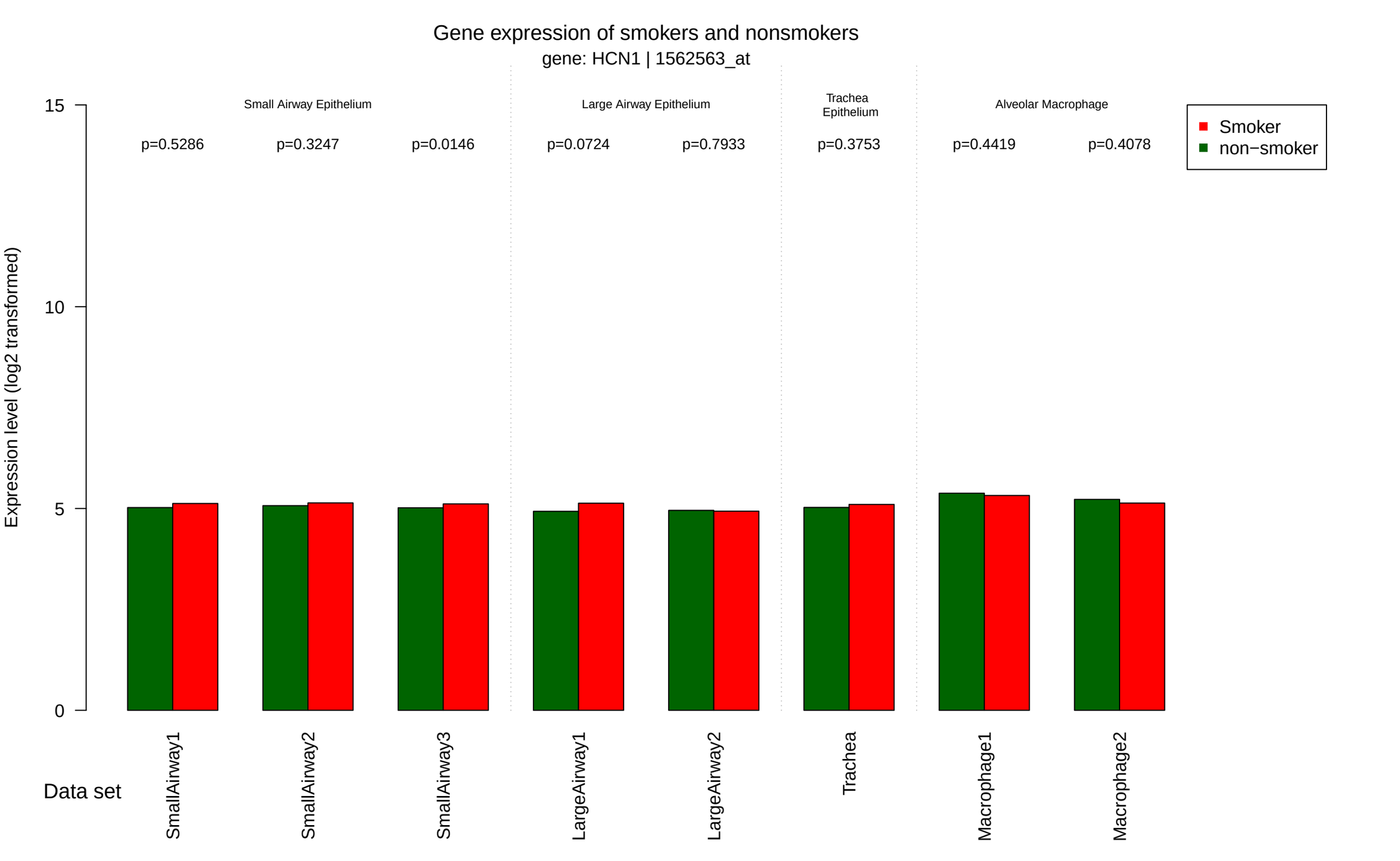
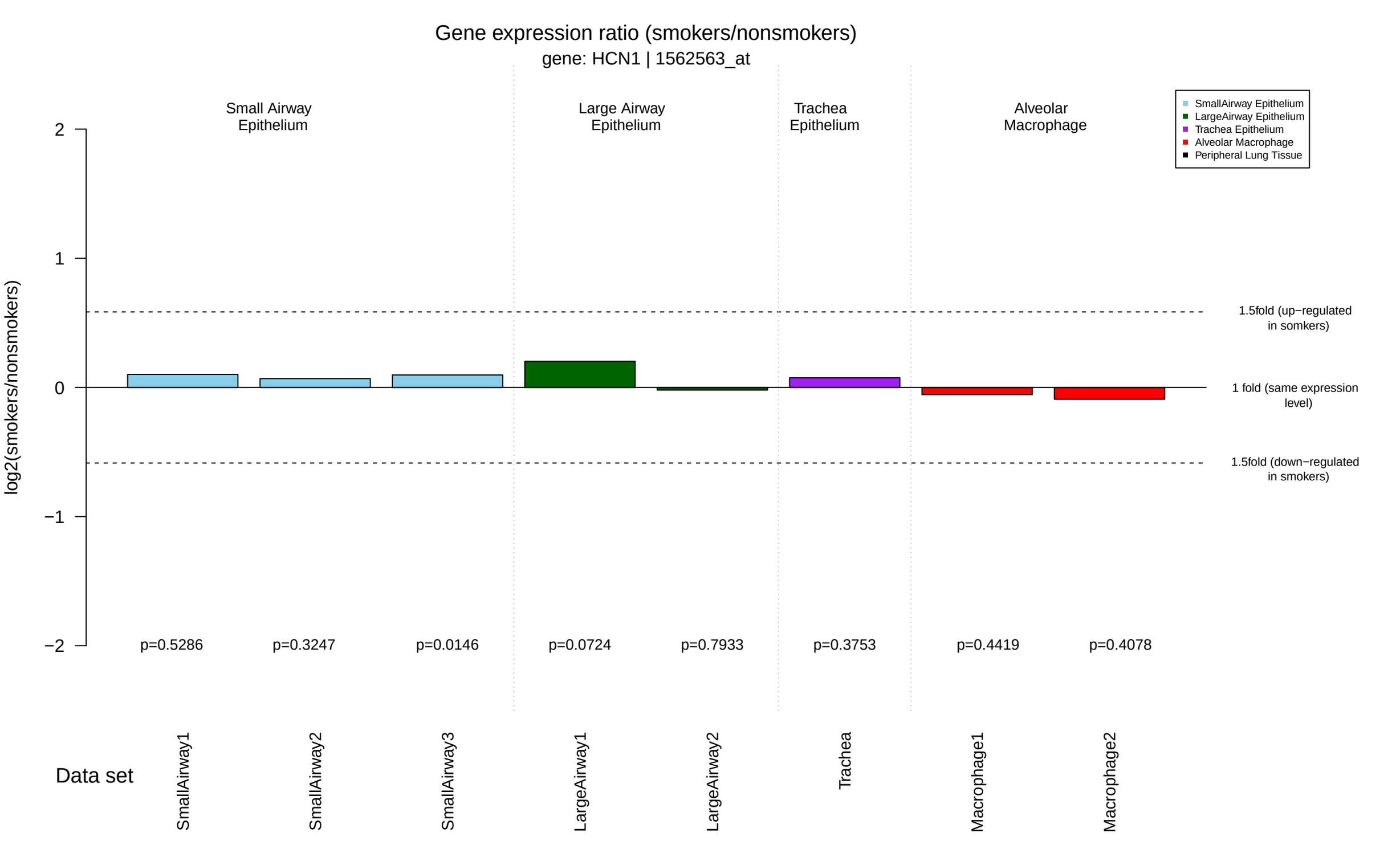
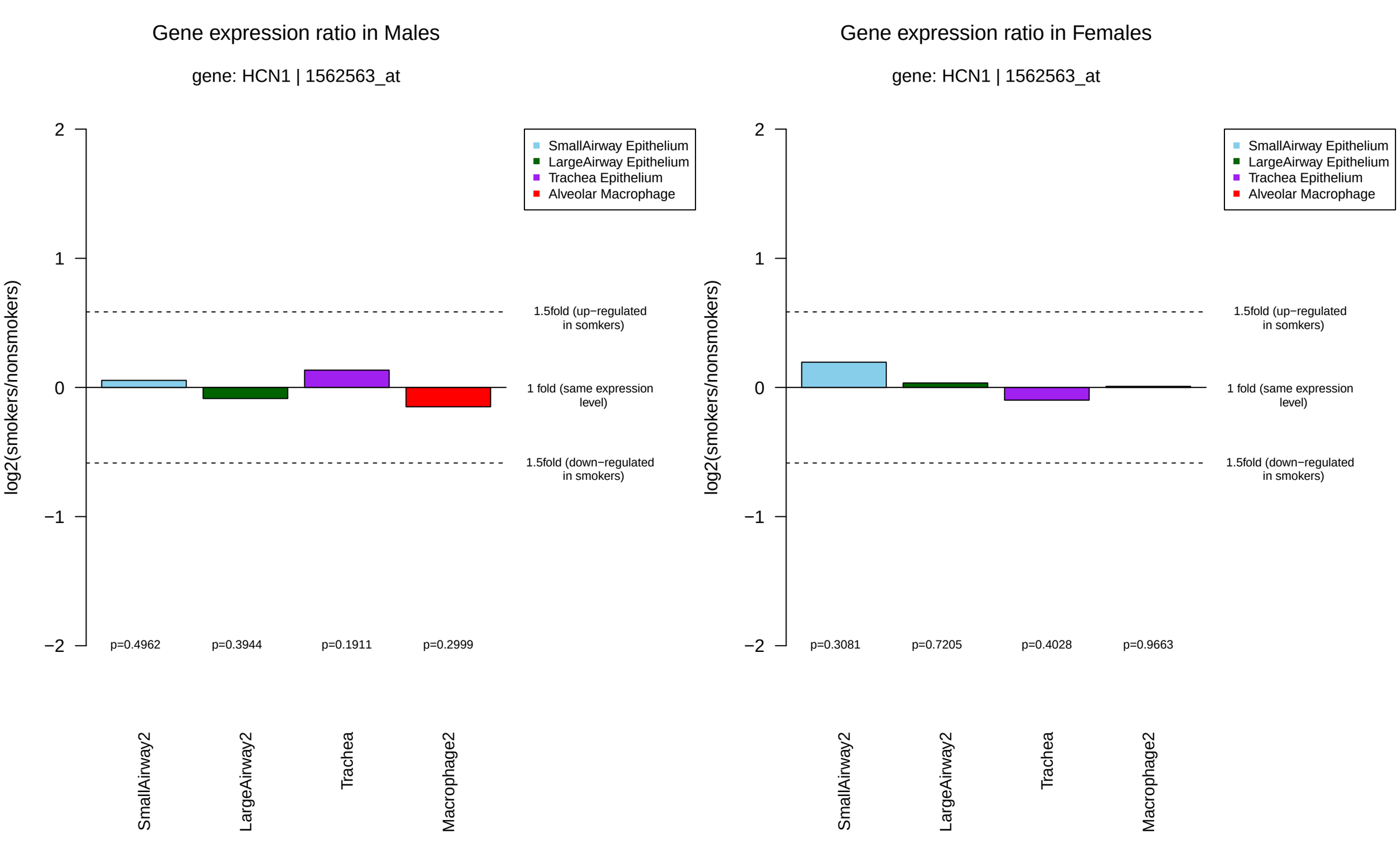
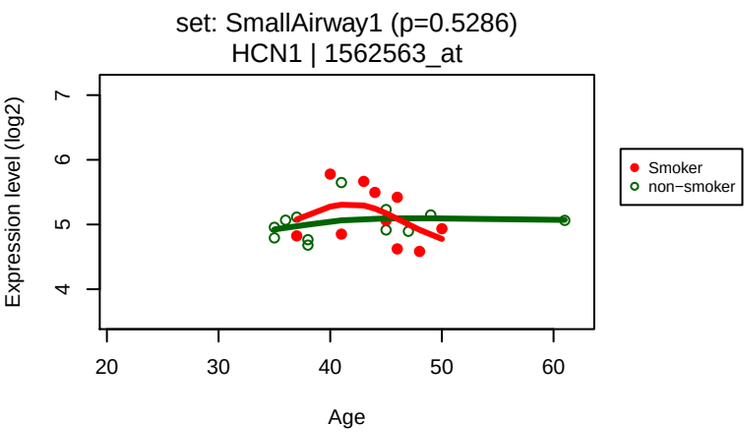
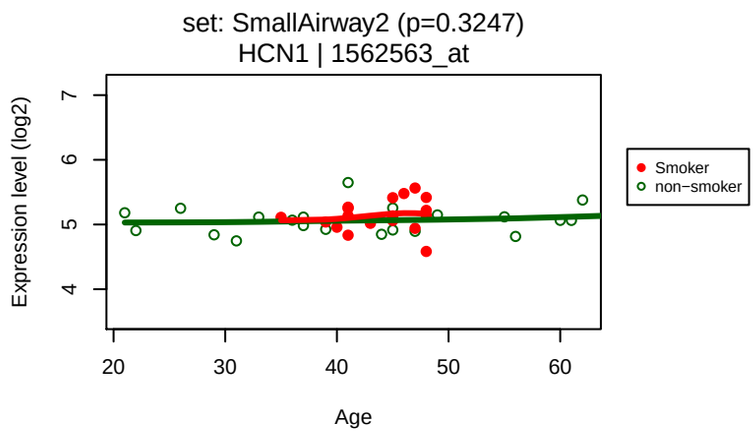
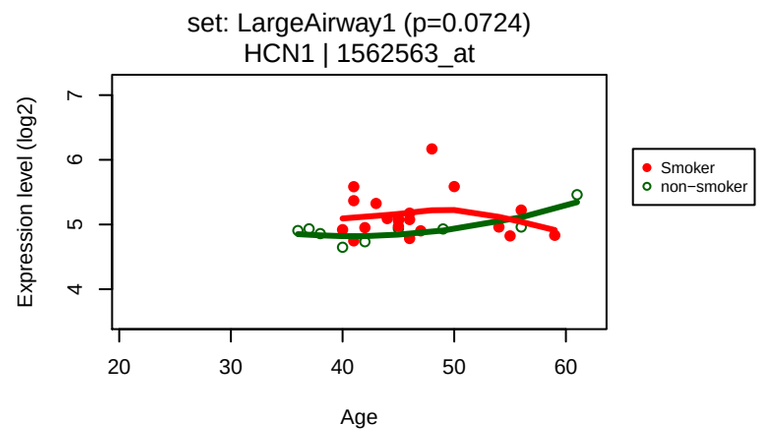

Gene expression of smokers and nonsmokers
gene: HCN1 | 1562563_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5286
p=0.3247
p=0.0146
p=0.0724
p=0.7933
p=0.3753
p=0.4419
p=0.4078
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

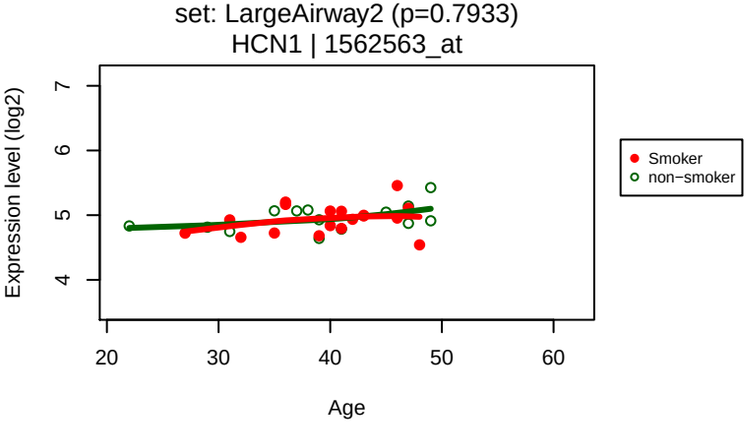

Gene expression ratio (smokers/nonsmokers)
gene: HCN1 | 1562563_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5286
p=0.3247
p=0.0146
p=0.0724
p=0.7933
p=0.3753
p=0.4419
p=0.4078
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

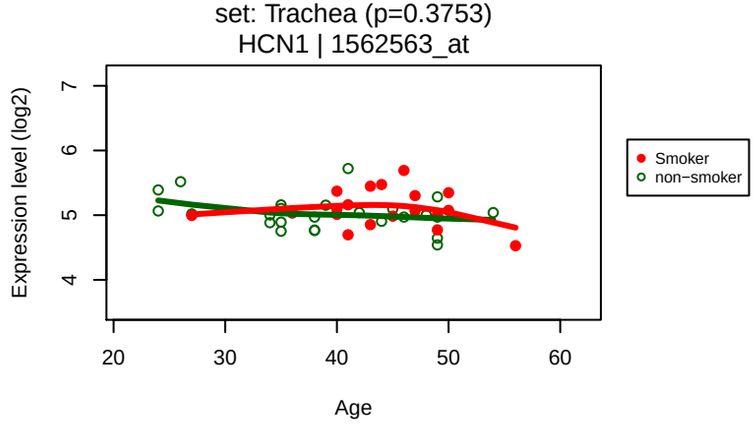

Gene expression ratio in Males
gene: HCN1 | 1562563_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4962
p=0.3944
p=0.1911
p=0.2999
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HCN1 | 1562563_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3081
p=0.7205
p=0.4028
p=0.9663
SmallAirway2
LargeAirway2
Trachea
Macrophage2

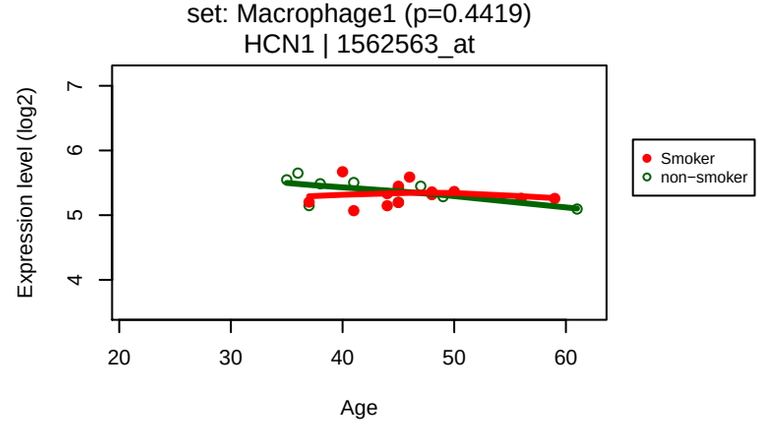

set: SmallAirway1 (p=0.5286)
HCN1 | 1562563_at
Expression level (log2)
Age
Smoker
non−smoker

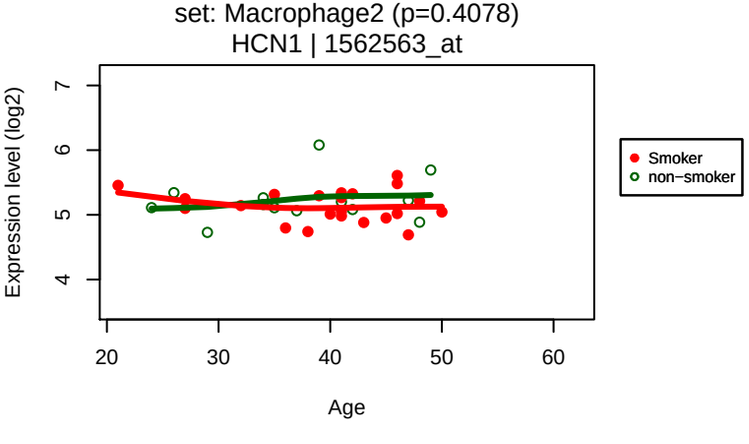

set: SmallAirway2 (p=0.3247)
HCN1 | 1562563_at
Expression level (log2)
Age
Smoker
non−smoker

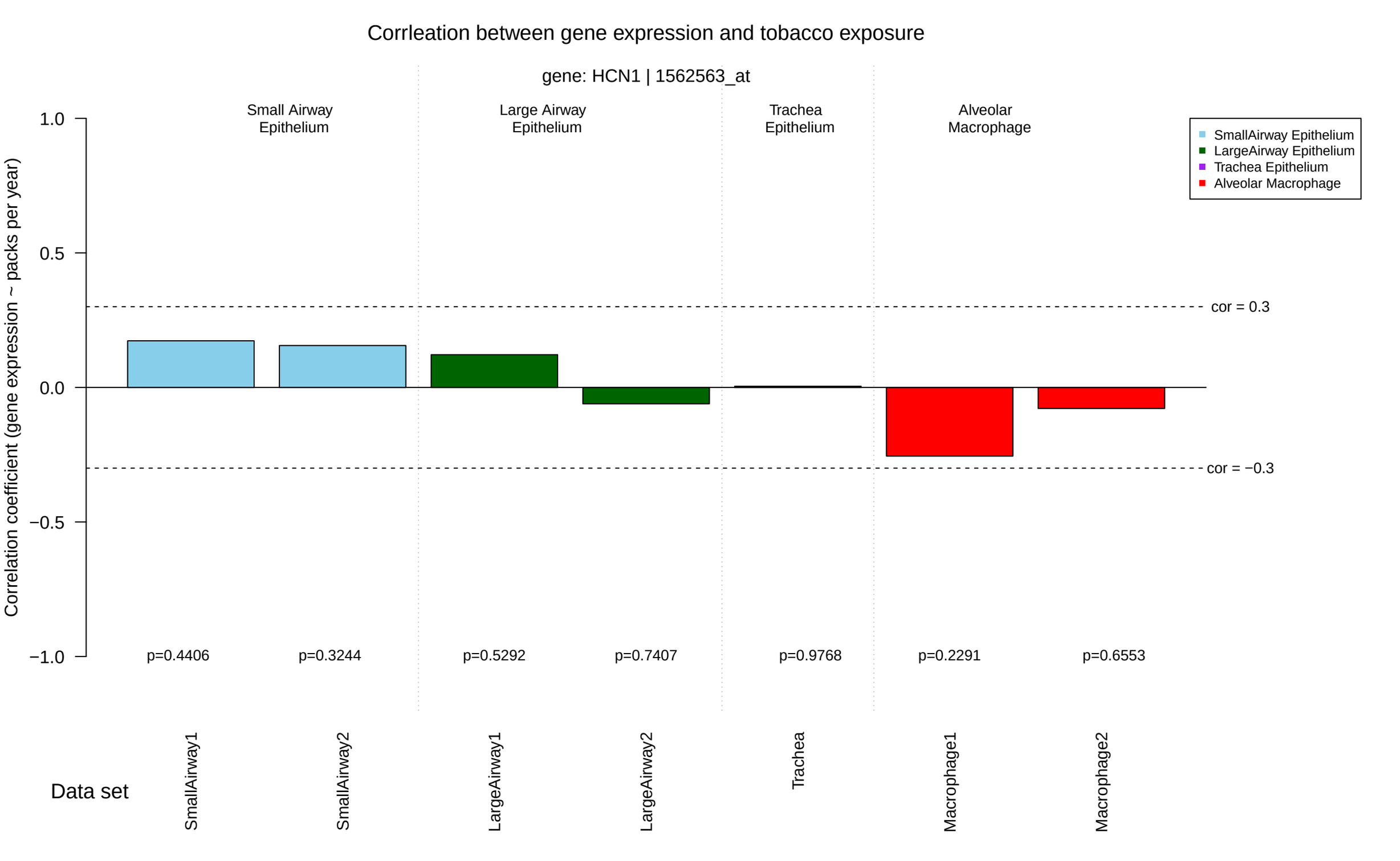

set: LargeAirway1 (p=0.0724)
HCN1 | 1562563_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7933)
HCN1 | 1562563_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3753)
HCN1 | 1562563_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4419)
HCN1 | 1562563_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4078)
HCN1 | 1562563_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HCN1 | 1562563_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4406
p=0.3244
p=0.5292
p=0.7407
p=0.9768
p=0.2291
p=0.6553
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2